Sivakorn Khundilokrattaya

3rd-Year Computer Engineering Student • Seeking Data Science, ML & Quantitative Fintech Internship

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EDUCATION & CREDENTIALS

Srinakharinwirot University (SWU) – B.Eng. in Computer Engineering 2024 – Present (Expected 2028)

- **Focus & Coursework:** Statistical ML, Credit Risk Modeling, Explainable AI; Machine Learning, Probability & Statistics, Database Systems, DSA.
- **NHK Spring Undergraduate Scholarship (2026–2027):** Corporate engineering scholarship (GPAX ≥ 2.75).
- **Standardized Assessments (2026):** KBTG Data Analyst STAMP (Score 97/150, Benchmark: 90) • Oxford Placement Test English Proficiency (CEFR B1, Score 51/120, AUA / Oxford University Press).

TECHNICAL SKILLS

ML & Explainable AI: LightGBM, CatBoost, XGBoost, Scikit-Learn, PyG, Optuna, OptBinning, TreeSHAP, DiCE

Data & Engineering: Python (Pandas, NumPy, SciPy), SQL (PostgreSQL), FastAPI, ONNX, Docker, React, Git

Languages: Thai (Native), English (CEFR B1 – Oxford Placement Test), Japanese (Elementary)

FEATURED DATA SCIENCE & ML PROJECTS

K-NEXUS – FlowSense & TrustGraph (Synthetic Cashflow & Risk Prototype) • App | Showcase 2026

PyG (R-GCN), LightGBM, ONNX, FastAPI • KBTG Kampus Hackathon 2026 – Top 30 Overall & Top 10 Data Science

- Evaluated pre-transfer scam-risk screening across 1,354 synthetic transactions, showing that a simpler tabular baseline (0.996 PR-AUC) outperformed a complex R-GCN + LightGBM graph model (0.994 PR-AUC).
- Prevented inaccurate 30-day cashflow forecasts by disabling an unvalidated ML module on synthetic data and serving an explainable rule-based spending plan in FastAPI and React.

Credit Risk Scoring & Explainable AI (XAI) Decision-Support Prototype • GitHub 2026

LightGBM, Optuna, TreeSHAP, FastAPI • Research Prototype

- Improved loan default prediction to 0.785 ROC-AUC and 43.2% KS separation (vs. 0.742 logistic baseline) on Home Credit data by tuning a 5-fold LightGBM model with Optuna and mapping to a 300–850 scorecard.
- Built an interactive loan-review dashboard in FastAPI and Next.js that explains rejection reasons with TreeSHAP and generates hypothetical approval adjustments using DiCE counterfactuals.

Shopee Smart COD Reliability & Adaptive Checkout Prototype (Synthetic Data) • GitHub 2026

Python, Scikit-Learn, FastAPI, JavaScript • Shopee Ultimate Case Challenge 2026 – Operations Strategy Track

- Predicted cash-on-delivery (COD) failure risk at 0.840 ROC-AUC on 3,500 held-out synthetic orders (from 50,000 simulated orders) using a Random Forest pipeline to flag high-risk checkouts before shipping.
- Modeled a simulated 3.58M THB/month return-cost savings scenario (assuming 58% fewer failed deliveries) by designing a 4-tier checkout policy (Grades A–D) and a mobile checkout simulator in FastAPI.

Long Overdue Debtor (LOD) Default Prediction Pipeline • GitHub 2025

CatBoost, LightGBM, Optuna, Scikit-Learn • Aihack Thailand 2025 – National Finalist (Team OEIA)

- Improved 12-month loan default ranking from 0.633 to 0.653 ROC-AUC on imbalanced consumer loan records (~8.3% default rate) by blending CatBoost and LightGBM predictions across 5-fold cross-validation.
- Captured borrower distress signals using engineered debt-to-income, credit utilization, and missingness features, and published a reproducible synthetic data pipeline to protect confidential competition records.